



**Cape Peninsula
University of Technology**

FACULTY OF INFORMATICS & DESIGN

Diploma in Information & Communication Technology: Applications Development

INFORMATION SYSTEMS 3 ITS362S

SUBJECT GUIDE:	2022
COURSE CODE:	ITS362S
NQF LEVEL:	6
PREREQUISITE SUBJECTS:	PREREQUISITE SUBJECTS: COURSE CODES: ADT262S, APD262S, ITS262S

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ORGANISATIONAL COMPONENT

1. INTRODUCTION

This subject will build on the basics of database modelling and development techniques acquired in the previous 2 years. Information Systems 3B also covers the development of databases for applications systems with an emphasis on problem solving. Advanced related theoretical concepts such as database implementation, testing and quality will also be addressed. This subject will require students to practice previously acquired programming, academic writing and communications skills.

Although the emphasis during this year is to acquire, interpret and discuss new knowledge, the critical evaluation of IT problems, the practical application thereof and the development of practical solutions will be encouraged through the use of real life business entity.

1.1 PURPOSE OF SUBJECT

To develop the necessary skills, which will enable the student to:

- ❖ Manage the development and implementation of a Database successfully
- ❖ Design, Develop, and implement a database using the Relational database model or the object-oriented database model.
- ❖ Analyze data using data warehousing techniques

General skills like problem solving, usage of technology, i.e. WorkBench, DBMS (MySQL), applying Internet ideas, collect, analyse and organise information when completing assignments or working on the end year project.

1.2 SAQA LEVEL DESCRIPTORS FOR NQF LEVEL 6:

1. **Scope of knowledge** – do you understand what you're studying?

Scope of knowledge, in respect of which a learner is able to **demonstrate**: detailed knowledge of the **main areas** of one or more fields, disciplines or practices, including an understanding of and the ability to **apply the key terms**, concepts, facts, principles, rules and theories of that field, discipline or practice to **unfamiliar but relevant contexts**; and knowledge of an area or areas of specialisation and how that knowledge relates to other fields, disciplines or practices

2. **Knowledge literacy** – can you apply that understanding?

Knowledge literacy, in respect of which a learner is able to **demonstrate an understanding of different forms of knowledge**, schools of thought and forms of explanation within an area of study, operation or practice, and **awareness of knowledge production processes**.

3. **Method & procedure** – can you apply specific theories, technique and methods of what you are studying?

Method and procedure, in respect of which a learner is able to demonstrate the ability to evaluate, select and apply appropriate methods, procedures or techniques in investigation or application processes within a defined context.

4. **Problem-solving** – can you think for yourself, critically, and find solutions to problems?

Problem solving, in respect of which a learner is able to **demonstrate the ability to identify, analyse and solve problems in unfamiliar contexts, gathering evidence and applying solutions** based on evidence and procedures appropriate to the field, discipline or practice.

5. **Ethics & professional practice** – can you stick to codes of conduct and do you understand the values of your society?

Ethics and professional practice, in respect of which a learner is able to demonstrate an **understanding of the ethical implications of decisions and actions** within an organisational or professional context, based on an awareness of the complexity of ethical dilemmas

6. **Accessing, processing & managing information** – can you interpret and analyse information relating to what you are studying?

Accessing, processing and managing information, in respect of which a learner is able to **demonstrate the ability to evaluate different sources of information**, to select information appropriate to the task, and to apply well-developed processes of analysis, synthesis and evaluation to that information.

7. **Producing & communicating information** – are you able to accurately communicate what you have learnt, in words and writing?

Producing and communicating information, in respect of which a learner is able to **demonstrate the ability to present and communicate complex information** reliably and coherently **using** appropriate academic and professional or occupational conventions, **formats and technologies** for a given context.

8. **Context & systems** – do you understand the systems and environment of your field of study?

Context and systems, in respect of which a learner is able to demonstrate the ability to **make decisions and act appropriately in familiar and new contexts**, demonstrating an understanding of the **relationships between systems**, and of how actions, ideas or developments in one system impact on other systems

9. **Management of learning** – are you able to evaluate and take responsibility for your own learning, as well as the learning of others?

Management of learning, in respect of which a learner is able to **demonstrate the ability to evaluate performance against given criteria**, and accurately identify and address his or her **task-specific learning** needs in a given context, and **to provide support to the learning** needs of others where appropriate.

10. **Accountability** – are you able to properly use resources, to be accountable and responsible for your actions, as well as work effectively?

Accountability, in respect of which a learner is able to demonstrate **the ability to work effectively in a team or group**, and to **take responsibility** for his or her decisions and actions and the decisions and actions of others within well-defined contexts, including the responsibility for the use of resources where appropriate.

1.3 LEARNING OUTCOMES AND ASSESSMENT CRITERIA NQF LEVEL 6:

Learning outcomes of subject	Associated assessment criteria
Consult the SAQA level descriptors and Blooms' taxonomy to define the learning outcomes to be achieved by students. On completion of this subject, students will be able to appraise aspects in Data Modeling, Data Administration, System Integration, namely:	Use the CPUT guidelines on how to write learning outcomes and associated assessment criteria. A learning outcome may have more than one assessment criterion. On the completion of the LO, the student will be able to appraise as LO, by being assessed doing the following:

1. To appraise the role of databases and database management systems in managing organizational data and information by using advanced conceptual data modelling techniques (Data Modelling).	• Develop at least one conceptual data modelling technique from those available (such as entity-relationship modelling) to capture the information requirements for an enterprise domain. • Communicate effectively the concepts of designing high-quality databases. • Maintain a portfolio of evidence on advanced data modelling <u>Examples:</u> • Create an ERD using CHEN and UML • Map ERD to physical database • Normalization and PDFDs of proposed design
2. To appraise simple database administration tasks by using data manipulation language statements (Data Administration).	• Implement the data definition, data manipulation, and data control language components of SQL in the context of one widely used implementation of the language. • Communicate effectively the concepts of database administration. • Maintain a portfolio of evidence on data administration <u>Examples</u> • Construct software solutions using MySQL/MariaDB/ACCESS/ Workbench or any other RDBMS as the development tool.
3. To appraise the basic mechanisms for accessing integrated relational databases from various types of application development environments (System Integration).	• Develop a software system which is integrated across different platforms. • Communicate effectively the concepts of system integration. • Maintain a portfolio of evidence on system integration <u>Examples</u> • Understand and Implement a data warehouse using a star schema or another technique. • Understand the issues involved in transaction management and concurrency control • Understand Big data and its components

Teaching & learning strategy for subject

Include details about the teaching-learning methods that will be adopted in this subject.

1. A blended learning approach will be adopted which is a combination of e-learning via the CPUT's Blackboard learning management system, and classroom lectures if COVID measures permit. Small groups will be accommodated on campus within the safety precautions imposed by government.
2. Assessment on campus to combat online assessment challenges (All covid-19 protocols will be strictly implemented).
3. Group learning will take place in groups allocated by the lecturer to promote learning from peers.
4. Formative assessment and summative assessment to determine the degree of attainment of learning outcomes.
5. Practical assessments will be conducted where students demonstrate their understanding on the different topics for this subject.

2. General2

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Co-ordinator		Engineering 2.23 (5)	(021) 460 3285	
Lecturers	W Rothman	Engineering 2.23 (11)	(021) 460-3285	
	Mrs Sheetal	Engineering 2.52A	(021) 460-3010	
Domain Coordinator	K Naidoo	Engineering 2.48	(021) 460-3274	naidook@cput.ac.za
Head of Department	B. Kabaso	Engineering 2.30	(021) 460-3713	kabasob@cput.ac.za
Secretary	N. Allie	Engineering 2.1	(021) 460-3010	alien@cput.ac.za
Administrative Assistant				
Support Technician	S. Salie	Engineering 1.4	(021) 460-4209	saliesh@cput.ac.za
Support Technician	R. Pietersen	Engineering 1.4	(021) 460-4209	pietersenr@cput.ac.za
Support Technician	I. Saliem	Engineering 1.28	(021) 460-3031	saliemi@cput.ac.za
Support Technician	C. Sindayigaya	Engineering 1.28	(021) 460-3031	sindayigayac@cput.ac.za

2.2 Class Times and Venues

As published on <http://myclassroom.cput.ac.za> and notice boards. The subject consists of two practical and two theory periods per week. **Time-table** (Usually 4 periods per week, now 1 hour on Monday @ 12h00)

2.3 Learner Management System (LMS)

<https://myclassroom.cput.ac.za/>: Blackboard used for general communications and announcements, posting of course material, assignment and announcements. It is the responsibility of the learner to check Blackboard regularly. BlackBoard will be used in conjunction with Moodle. Please follow link on institutional homepage.

Most importantly students esp. in the third year MUST stay in contact with the lecturer at all times and promptly acknowledge any emails or requests from the lecturer. The student is solely responsible to update him/herself with any communication from the lecturer/ department or faculty. BlackBoard and Moodle will be used as the main vehicle for communication and it is the student's responsibility to maintain communication at all times.

Please note that it is wise to do a direct connection with BlackBoard seeing that there may be intermittent connection using <https://myclassroom.cput.ac.za>

3. STUDY MATERIALS

3.1 Prescribed Textbook

TEXTBOOK:	Database Principles 10e authors are Coronel, Morris, Rob ISBN 1-133-31197-0 Cengage International Edition 10th Ed. 2013 <u>OR</u> Rob P and Coronel C : "Database Systems" Cengage International Edition 2008 Available as eBook at Van Shaik Bookstore on campus
	LEARNING MATERIALS
RECOMMENDED BOOK/S:	Database Principles: Fundamentals of Design, Implementation, and Management. Second Edition, 2018. Carlos Coronel, Steven Morris, Peter Rob, Keeley Crockett. ISBN: 978-1-4080-4863-4
WEBSITE/S:	http://wrru.co.za/moodle

OTHER MATERIALS:	See BlackBoard or Shared Drive for Web site
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3.2 Recommended Readings

3.3 Websites

- www.bookboon.com
- <https://iacademy.oracle.com>
- <https://www.mysql.com/products/workbench/>

4. MODULE CREDITS

This module has 20 credits on NQF level 6. Please refer to the Faculty prospectus on <http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus> for further information.

4.1 Planning of Time Allocation for Learning Activities

There are 200 notional hours allocated to this module. These notional hours comprise of 60 hours (30%) for contact time and the remainder will be self-study and other tasks, assignments, and tests.

This module has two theory assessments, one research essay and two practical assessments. Term 1 is Theory Test (20 %), Assessment 2 will also be a theory Test (30%), Assessment 3 is a project using SQL (20%) and Assessment 4 is a theory test and eTests (30%). Please note that the practical part MySQL Workbench will be 20% of the year mark.

20	30	20	30	100
T1	T2	T3	T4	

METHOD	% OF YOUR LEARNING TIME
Seminars / Practicals (face-to-face, limited interaction or technologically mediated)	30%
Independent self-study of standard texts and references and specially prepared materials (study guides, books, journals, using online tutorials etc.)	20%
Independent self-study, time needed to prepare for assessment (e.g. tests/exams)	25%
Student learning groups (working with your peers for group assignments)	25%

5. ETHICAL CONDUCT

5.1 Roles and Responsibilities

Kindly refer to CPUT's Policy Docs and Student Guidelines:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.2 Class attendance and participation

Kindly refer to page 18 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.3 Grievance procedures

Kindly refer to page 61 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

Create a CRIMs ticket when it necessitates action.

5.4 Plagiarism

Kindly refer to page 30 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

*NB: An amended version of the General Handbook will be emailed to you during the first term.

6. ASSESSMENT

6.1 Assessment policy and regulations

Assessment policy and regulations

- Pass requirements will be 50 % on the whole.
- Late submission of assignments WILL NOT be marked or considered, due to the rolling nature of the subject.
- Any absence from a test must be accompanied by a doctor's letter. The doctor's letter must be handed in not later than one week after the test was written. Any late submissions of such letters will not be considered after this period. Any other excuses or reasons for not writing a test can be taken up with the HOD.
- Any mark disputes must also be sorted before the next test/assessment takes place. Any assessment marks/grading will be considered final and binding once the next assessment is taken down.
- Plagiarism refers to the appropriation of the work or ideas of others. Plagiarism is both unethical and illegal and may be regarded as a criminal offence in terms of the Copyright Act 98 of 1978. CPUT places a high premium on its academic standards and subscribes to a value system that requires strong action against plagiarism. Being regarded as a serious contravention of the University's rules, plagiarism can lead to expulsion from the University. Any student who makes himself/herself guilty of such an offence will be dealt with according to the rules of the faculty. Ensure you steer clear of such atrocities.

- THE ROTHMAN EQUALIZER OR LIQUADIZER:

If you obtain 50 % or greater than (difference in percentage) for your theory test when looking at the difference between your practical and theory test, your practical will change to that lower percentage as well. E.g. - you are working two on a project and one person scores well in the test and the other gets 50 % or greater difference in percentage regarding theory and practical you will have to live with the theory mark for the practical as well whether you obtained 90 % for the practical. The person satisfying the condition will keep his/her mark.

5.2 Assessment opportunities: administration

Different assessment types are used in ITS362S to gauge the extent of what you have learnt from attending classes and engaging with course content.

- The student will be expected to demonstrate his/her understanding of course related theory or practical in either theoretical and/or practical manner. Due/owing to continuous evaluation, no examination is written, but assessments will be taken down continuously. This has however a major implication, which puts a great responsibility on the student to work hard throughout the year. With an exam the focus is on the last assessment, but with continuous evaluation the focus is throughout the year. Having bad marks initially cannot be rectified in the end. The student must constantly monitor his/her performance during the year and work accordingly. My suggestion is to keep a worksheet with your assessments to indicate the attention ITS362S needs during your year of study.
- Some of these assessments will be awarded marks, while others are there to provide the student with feedback. The feedback will enable the student to monitor his/her development during the course.
- Assessments will take the form of written tests, assignments, essays and practical demonstrations of your understanding on the different topics for this subject. The content of this subject varies from theory to practical to writing skills.
- The assessments will also more often be individual assessments, but group assessments are also evaluated.
- Most importantly is that the student should not lose sight of the fact that his/her studies is still paramount even in the last term of the year.

After certain sections of the work are completed, the student will be expected to complete some assessment (prac or theory) to finalize the work. These assessments are unfortunately not repeated or taken down again and the learner only has one chance to complete this. The results will count towards a certain (usually 5 %) percentage (see table below). These additional assessments will have formal dates and timeslots as announced beforehand.

20	30	20	30	100
T1	T2	T3	T4	
Test/E- Work	Test/E- work	Test /E- Work	Tes/E- Work	

6.2 Assessment Opportunities: Administration

There will be one test per term. There will be a sick test in June for the first semester formal assessments and a sick test in November for the second semester formal assessments.

ASSESSMENT BREAKDOWN AND WEIGHTS

ASSESSMENT CRITERIA	1 ST TERM	2 ND TERM	3 RD TERM	4 TH TERM	TOTALS
Nature of assessment	W/M	W/M	W/M	W/M	
Will assessment be included in this term?	Yes	Yes	Yes	Yes	
Total assessment weight for the term	20%	30%	20%	30%	100%
Will the assessment be moderated?	No	No	No	Yes	

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STUDY COMPONENT

7. SUBJECT SPECIFICATIONS

7.1 Purpose of the Subject

The coverage of this subject is limited to:

- Database Design and Application Development
- Object-Oriented Database Design, and Relational Database Design
- Data Warehousing

PRE- AND CO- REQUISITE KNOWLEDGE

It is assumed that following learning is in place:

- Knowledge of Relational Models
 - Data Modeling
 - Object-Oriented Concepts
- Database basics as done in Year 2 of study

SUBJECT GOALS AND OBJECTIVES

On successful completion of this subject, the student will be able to:

1. Develop software solutions using MS ACCESS/MYSQL/Workbench or any other RDBMS as the development tool.
2. Understand and use theoretical concepts relating to Database Design, Processing and Management, while developing a database system according to the Systems Development Life Cycle (SDLC) and Database Life Cycle.
3. Develop and apply information literacy skills and strategies within an academic context. You will be able to use information to help answer an essay question, by being able to analyse and evaluate the value of information that you find, and then communicate and structure this information into a suitable written form like an essay.

4. Use your understanding of database design, in the conceptualisation, planning and design of a database.
5. Understand and use the core principles and practices associated with the concepts of design for software development as well as the creation of a relational database.
6. Understand the issues involved in transaction management
7. Understand data warehousing techniques and implement a data warehouse using a star schema or another technique
8. Apply basic approaches and strategies of project management to define, plan, design develop and implement an IT related project with a substantial database element in conjunction with another subject such as Internet Programming.
7. Use an understanding of basic technical writing conventions to create and develop original content in a project scoping document.
8. Develop a range of general abilities like using information appropriately, communicating your ideas and thoughts in written, oral and visual forms, managing and organising yourself effectively so that you can complete your study tasks effectively and on time, work together with other people in a partnership or in groups where you are able to listen to others and solve problems collectively.
9. Use a design tool to generate design and development solutions
10. Work as a team member on an IT project
11. Research and evaluate an IT problem and provide recommendations from your research

TOPIC SEQUENCING

Term 1

The following are the main topics in the Information Systems 362 (Database)

CURRICULUM DETAILS

Topic 1	Conceptual Database Design using advanced ER diagrams and Normalization
Topic 2	Relational Algebra
Topic 3	Database implementation and application development using SQL
Topic 4	Distributed Database Systems and Concurrency Control
Topic 5	Data Warehousing, multidimensional databases and Decision Support Systems
Topic 6	Big Data

Theoretical Assessment: 70% Theory + 20% Prac and 10% e-Tests/Assignments

Test 1: 20 % (Theory assessments & Prac Assign and Elec Class Tests) March

Test 2: 30 % (30% Theory + 10% eTests/e-Work) June

Test 3: 20 % (30% Prac and + 10 % Small Projects/E-Tests/E-Work)

Test 4: 30 % (e-Tests/E-Work)

Practical Assess: 20 % Includes E-Work and/or Practical Assignments

Note: The academic structure will move upwards and this is just a guideline which will be determined by the pace of the class.

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7.2 Module structure

TERM 1: 31 JANUARY– 18th MARCH – 7 WEEKS

Chapter No.:	Theme / Topic	Specific Outcomes
TERM 1	Design Concepts	
W1 31/01	Ch 4, 5,6	ERD Revision
W2 07/02	Ch 4, 5,6	ERD/Normalization Revision
W3 14/02	Ch 4, 5,6	Normalization Revision
W4 21/02	Ch 5	Normalization/Advanced Modelling
W5 28/02	Ch 3	Advanced Modelling
W6 07/03	Ch 7 or Catchup	SQL/Catchup
W7 14/03	Revision/Assessment Week	
	TERM 2: 28th MARCH – 27th JUNE- 13 WEEKS	
TERM 2	Advanced Design and Implementation	
W1 28/03	Ch 7	SQL Basics
W2 04/04	Ch 8	Advanced SQL
W3 11/04	Ch 8	Advanced SQL
W4 18/04	Ch 9	Database Design - Overview
W5 25/04	Ch 10	Transaction Management
W6 03/05 (02/05 HOLIDAY)	Ch 10	Transaction Management
W7 09/05	Revision	
W8 16/05	Project Week	
W9 23/05	Assessments	
W10 30/05	Assessments	
W11 06/06	Deferred Assessments	
W12 13/06		
W13 20/06		

	TERM 3: 18th JULY – 09th SEPTEMBER	7 Weeks
TERM 3	Advanced Database Management Concepts	
W1: 18/07	CH 12	Distributed DBMSs
W2: 25/07	CH 12	Distributed DBMSs
W3: 01/08	CH 12	Distributed DBMSs
W4: 08/08	Ch 13	Business Intelligence and Data Warehouses
W5: 15/08	Ch 13	Business Intelligence and Data Warehouses
W6: 22/08	Ch 13	Business Intelligence and Data Warehouses/Big Data
W7: 29/08	Ch 14	Big Data Analytics and NoSQL
W1 05/09	Ch 14	Big Data Analytics and NoSQL
	TERM 4: 19th SEPTEMBER – 15th DECEMBER	7 Weeks
TERM 4	ADVANCED DATABASES	DATABASES
W1 19/09	Ch 15	Database connectivity and Web Technologies
W2 26/09	Ch 15	Database connectivity and Web Technologies
W3 03/10	Ch 16	Database Administration and Security
W4 10/10	Revision/Catchup	Catchup
W5 17/10	Revision	
W6 24/10	ASSESSMENTS	
W7 31/10	ASSESSMENTS	
Appendix C		
Appendix D	Connecting to MYSQL Server	
Appendix F		

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OUTCOMES OF THE DATABASE PROGRAMMING PAT

SHOW PROFICIENCY IN: Application of Database Design and Implementation

PERIOD ENDING:	01 March - 31 March	12 April – 25 June	15 July - 03 September	13 Sept – 15 December
TERM:	1 st Term (5)	2 nd Term (9+2)	3 rd Term (5+2)	4 th Term (3+2)
TEST WEEK	None	14 June – 21 June	23 Aug – 30 Aug	
PROJECT WEEKS		03 Jun – 21 Jun	TBA	TBA
MARKS DUE:	End June	End June	End Sept	End Nov
MARKS REVIEW:	TBA	TBA	TBA	TBA
FINAL MARKS:	TBA	TBA	TBA	TBA
ASSESSMENT SELECTION:	Yes	Yes	Yes	WIL
WEIGHT IN %:	(T1:20(15+5) + T2:30(25+5) + T3:(10+10) + T4: 30 (25+5) = 100			
MODERATION SELECTION:		Yes		Yes

BOOK:

TEXTBOOK:	Database Principles 10e authors are Coronel, Morris, Rob ISBN 1-133-31197-0 Cengage International Edition 10th Ed. 2013 <u>OR</u> Rob P and Coronel C : "Database Systems" Cengage International Edition 2008 Available as eBook at Van Shaik Bookstore on campus
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HOLIDAYS – NO LECTURES

Human Rights Day	21 March	University Holiday	15 June
Good Friday Day	02 April	Womens Day	09 August
Family Day	05 April	Heritage Day	24 September
Freedom Day	27 April	Day of Reconciliation	16 December
Workers Day	01 May	Christmas Day	25 December
Youth Day	16 June	Day of Goodwill	26 December

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7.3 Articulation with other subjects in the programme

This subject is linked to concepts taught in Information Systems II.

A great deal of database design, systems analysis, project management, programming etc. form part of the subject. Information Systems 3b teaches students to analyse, design and implement IT business applications.

2nd year Linkages

Information Systems Databases continues, expands and focuses on the design and application of databases, and concurrency, data warehouse and distributed database theory as well as the Software development life cycle (SDLC) which were introduced in the following second year subjects.

7.4 Learning presumed to be in place

As stated above, all aspects of IT come into play, but above all creativity plays an important role in applying these skills in the Databases realm. Furthermore:

- Knowledge of an information system and how and where it is used
- Understanding of basic database and database modelling, with Entity Relationship Diagrams and normalization
- Understanding the basics of query languages, especially SQL
- Ability to apply basic programming language(s)
- Problem solving skills
- Design skills
- Testing skills
- Reading and writing skills

7.5 Critical Cross-field Outcomes

The critical cross-field outcomes include, but are not limited to:

- Identifying and solving problems by using critical and creative thinking.
- Working effectively with others as a member of a team, group, organisation, community.
- Organising and managing oneself and one's activities responsibly and effectively.
- Collecting, analysing, organising and critically evaluating information.
- Communicating effectively using visual, mathematical and/or language skills in the modes of oral and/or written persuasion.
- Using science and technology responsibly, effectively and critically, showing responsibility towards the environment and health of others.
- Demonstrating an understanding of the world as a set of related systems by recognising that problem-solving contexts do not exist in isolation.

- Contributing to the full personal development of each learner and the social and economic development of society at large by making it the underlying intention of any programme of learning to make an individual aware of the importance of:
 - reflecting on and exploring a variety of strategies to learn more effectively;
 - participating as responsible citizens in the lives of local, national and global communities;
 - being culturally and aesthetically sensitive across a range of social contexts;
 - exploring education and career opportunities; and
 - developing entrepreneurial opportunities.

CRITICAL CROSS-FIELD OUTCOMES	DESCRIPTION
CCFO1	Identify and solve problems with responses which display responsible decisions, using critical thinking, have been made.
CCFO2	Work effectively with others as a member of a team, group, organisation or community.
CCFO3	Organise and manage oneself and one's activities responsibly and effectively.
CCFO4	Collect, analyse, organise and critically evaluate information.
CCFO5	Communicate effectively, using visual, mathematical and/or language skills in the modes of written and/or oral presentation.
CCFO6	Use science and technology effectively and critically, showing responsibility towards the environment and the health of others.
CCFO7	Demonstrate an understanding of the world as a set of related systems by recognising that problem-solving contexts do not exist in isolation.
CCFO8	Contribute to the full personal development of each learner and the social and economic development of the society at large by making it the underlying intention of any programme of learning to create awareness in individuals about the importance of: i) reflecting on and exploring a variety of strategies to learn more effectively; ii) participating as responsible citizens in the life of local, national and global communities; and iii) being culturally and aesthetically sensitive across a range of social contexts. (RSASAQA, 2001:24)

8. STUDY UNITS/STUDY THEMES

8.1 Study Units

<i>Study unit outcomes</i>	<i>Teaching events</i>	<i>Learning activities</i>	<i>Assessments</i>
Entity Relationship Diagrams (ERDs) and Extended ERDs	Lectures, practical work, group and individual class exercises	Use ER diagrams, revised with introduction to advanced ER features like specialization, aggregation and generalization	Tutorials Test Project work
ERD to database structure	Lectures, group and individual class exercises	Practical Conversion of an ER model into a database structure, Gain understanding of general rules governing relationships among tables/classes Practical - From design to implementation	Test Tutorials Project work
Advanced Data Modelling	Lectures, group and individual class exercises	Apply constructivist approach	Tutorials and Tests
Normalization	Lectures, group and individual class exercises	Apply constructivist approach	Tutorials and Tests
Introduction to SQL	Lectures, group and individual class exercises	Apply constructivist approach	Tutorials and Tests
Advanced Design and Implementation	Lectures, group and individual class exercises	Apply constructivist approach	Tutorials and Tests
Advanced SQL	Lectures, group and individual class exercises	Apply constructivist approach	Tutorials and Tests
Transaction Processing	Lectures and individual class exercises	Understand Transaction Management and Concurrency Control.	Tutorials Test
Database Performance Tuning and Query Optimization	Lectures and individual class exercises	Understand Transaction Management and Concurrency Control.	Tutorials Test
Distributed Databases	Lectures and individual class exercises	Gain understanding of the Distributed Database and distributed transactions and	Tutorials Test

		associated challenges.	
Business Intelligence and Data Warehousing	Lectures, case study and projects to be implemented in practical classes	The Need for Data Analysis, The Data Warehouse, Decision Support Systems, OLAP and data cubes	Tutorials, Test
Multidimensional Databases	Lectures, case study and projects to be implemented in practical classes	Star Schema, From Problem to Design, Practical Assignments - Star Schema Design Problem	Tutorials, Project, implementation of Star Schema and Tests

8.2 Assessment Opportunities and Assessment strategy

Each class will be regarded as a tutorial and the outcome for each tutorial will be assessed and can be used as a mark to contribute to assessments. Please consult the table in the next sub-section to guide you regarding opportunities to earn marks for Databases.

8.3 Subject Outcomes and Associated Assessment Criteria

ASSESSMENT OVERVIEW FOR THE YEAR WITH MARK ALLOCATIONS		Weight
Internet Programming Theory/Prac Component		100 %
ASSESSMENT 1:	Summative test (Practical/Theory) (20%)	20%
ASSESSMENT 2:	Summative test (Practical/Theory) (30%)	30 %
ASSESSMENT 3:	Summative test (Practical/Theory) (20%)	20%
ASSESSMENT 4:	Summative test (Practical/Theory) (30%)	30%

TYPES OF ASSESSMENT AND WHY IT IS USEFUL FOR YOUR LEARNING

Different assessment types are used in this Subject to gauge the extent of what you have learnt from attending classes and engaging with course content.

- The student will be expected to demonstrate his/her understanding of course related theory or practical in either theoretical or practical manner. Due/owing to continuous evaluation, no examination is written, but evaluations will be taken down continuously as assessments. This has however a major implication, which puts a great responsibility on the student to work hard throughout the year. With an exam the focus is on the last assessment, but with continuous evaluation the focus is throughout the year. Having bad marks initially cannot be rectified in the end. The student must constantly monitor his/her performance during the year and work accordingly.
- Some of these assessments will be awarded marks, while others are there to provide the student with feedback. The feedback will enable the student to monitor his/her development during the course.
- Assessments will take the form of written tests, assignments, essays and practical demonstrations of your understanding on the different topics as mentioned above. The content of this subject varies from theory to practical to writing skills.
- The assessments will also more often be individual assessments, but group assessments are also evaluated as with a maximum of TWO students per group.
- After certain sections of the work are completed, the student will be expected to complete some assessment (prac or theory) to finalize the work. These assessments are *unfortunately not repeated or taken down again* and the learner only has one chance to complete this. The results will count towards a certain (usually 20 %) percentage. These additional assessments will have formal dates and timeslots as announced beforehand and completed through BB or Moodle.

8.4 Self-study Activities

RESOURCES	
Databases	MYSQL
	MYSQL Workbench
PREREQUISITES	
	All Tools must be installed on your home computer or Laptop
	WebServers, (Apache/IIS), PHP
	MYSQL WORKBENCH – Flipped Classroom on Youtube.
	ER Diagramming

9. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY
STUDENT ACKNOWLEDGEMENT
2021
DIPICTA: INFORMATION TECHNOLOGY
APPLICATION DEVELOPMENT INFORMATION SYSTEMS 3

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **APPLICATION DEVELOPMENT INFORMATION SYSTEMS 3** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname:
 (Print)

Student Number:

Contact number: (home/res) Contact number: (cell)

Signature:

Date: